

Surface roughness Ra: N grades, processes and values to specify

Roughness grades N and Ra values

The 12 N grades and their Ra values:

Grade	Ra (µm)	Ra (µin)
N1	0.025	1
N2	0.05	2
N3	0.1	4
N4	0.2	8
N5	0.4	16
N6	0.8	31
N7	1.6	63
N8	3.2	126
N9	6.3	248
N10	12.5	492
N11	25	984
N12	50	1968

1. Which Ra each process can hold

Common Ra ranges (µm) in production. These are ballpark figures: tool, material, speeds and machine easily shift them. It's what a shop achieves without special effort.

Process	Ra (µm)
Flame cutting, sawing	12.5 - 50
Sand casting	12.5 - 25
Drilling	1.6 - 6.3
Turning, milling	0.8 - 6.3
EDM (wire or sinker)	0.8 - 6.3
Reaming	0.8 - 3.2
Grinding	0.1 - 1.6
Honing	0.1 - 0.8
Polishing, superfinishing	0.025 - 0.4

Asking for 0.4 on a milled part means adding an operation (grinding or polishing). If that's not intended, it's over-specification you pay for on every part.

2. Which Ra to specify for each function

Ballpark values commonly used in general engineering. For bearings and seals, the manufacturer's documentation sets the exact value: it takes precedence.

Surface function	Ra (µm)
Non-functional surfaces, as-machined	6.3 - 12.5
Seating faces, fixed contact faces	3.2

Surface function	Ra (μm)
Common clearance fits (H7/g6, H8/f7)	0.8 – 1.6
Bearing seats, interference fits	0.4 – 1.6
Dynamic lip-seal counterfaces	0.2 – 0.8
Precision sliding surfaces, cylinder rods	0.1 – 0.4

Roughness must stay consistent with the tolerance: a dimension toleranced to a few micrometres on an Ra 6.3 surface can be neither made nor measured.

Reference values taken from the article <https://www.baouch.fr/en/blog/rugosite-ra-tableau-classes-procedes/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.